

Exponential Mayhem

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A mathematician created a calculator with buttons for 10^x , e^x , $\log x$, $\ln x$, and $\lceil x \rceil$ ($\lceil x \rceil$ denotes the least integer that is greater than or equal to x). One day, she was wondering how to compute different numbers using her calculator. How would you express the numbers from 2 to 16 using the available functions?

The Rules: The calculator's screen initially displays 0. Upon being pressed, each button (function) takes the value currently on the screen as its input.

The Puzzle: Assuming that the calculator can calculate logarithmic and exponential functions with infinite precision, your task is to create expressions that equate to every integer between 2 and 16, inclusive. For example, $3 = \lceil e^{e^0} \rceil$.